

An Experimental Evaluation of TCP Congestion Control Algorithms

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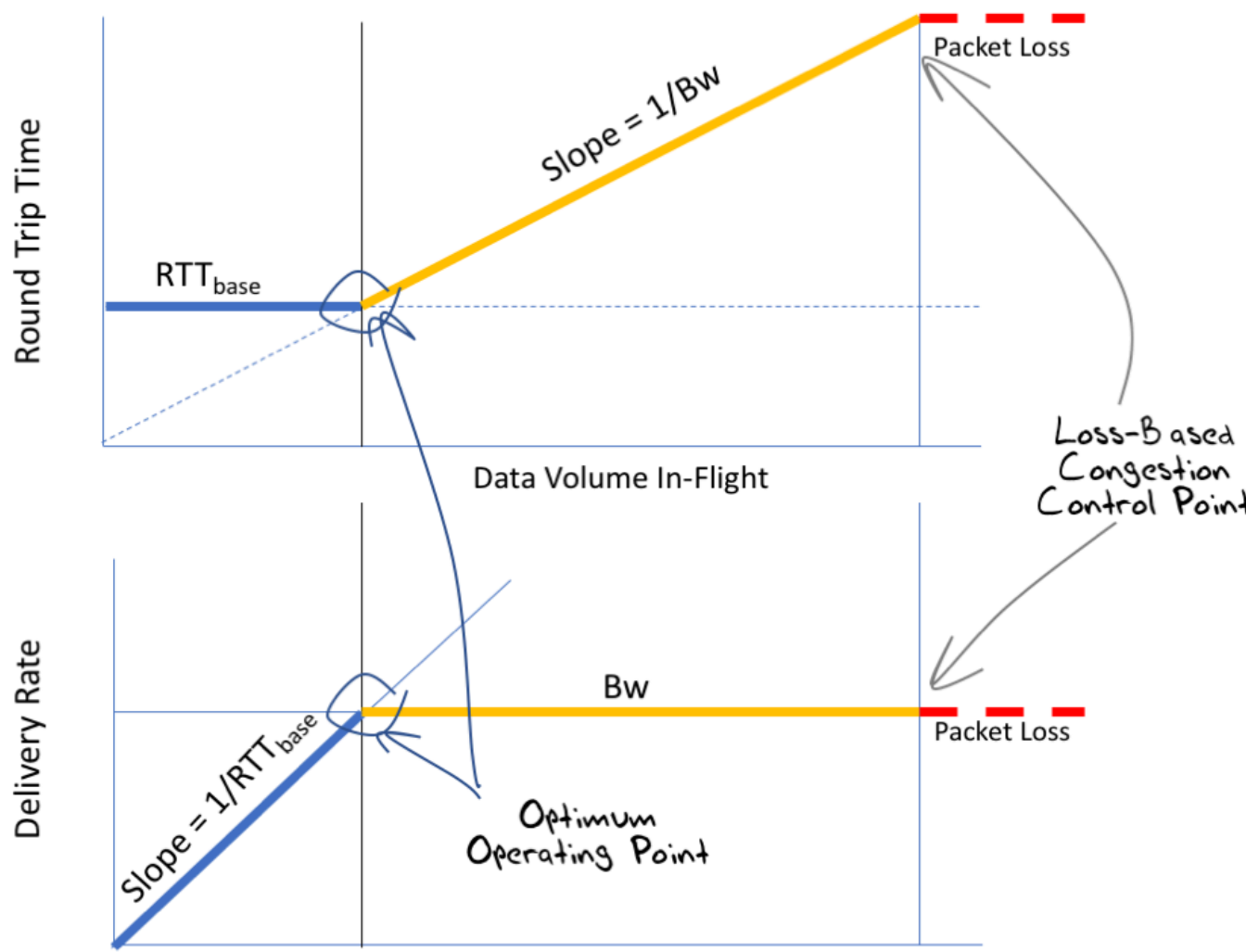
BBR Algorithm

Model-Based: Unlike Reno/CUBIC, BBR build: an explicit model of the network path.

BDP Targeting: Tries to send just enough data to fully use the connection without overloading.

Loss Resilience: Ignores random packet loss, preventing unnecessary throughput backoff.

Efficiency: Minimizes queueing delay by not filling buffers at the bottleneck.

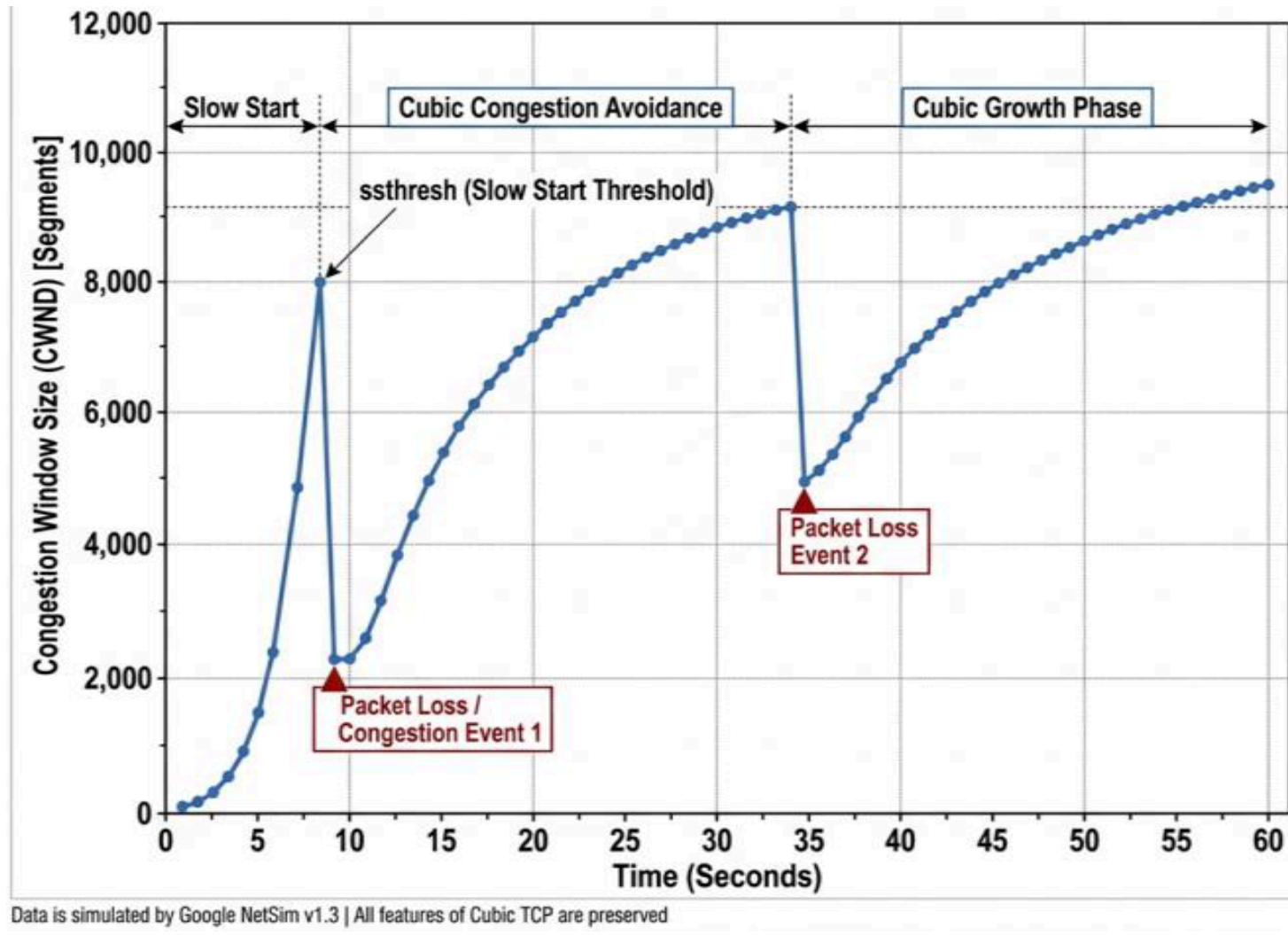


Cubic Algorithm

Loss-Based: CUBIC relies on packet loss as the primary signal for network congestion.

Cubic Function: Uses a cubic function of time since the last congestion event to scale the window.

Stability: Designed to maintain stability and fairness in high-bandwidth, high-latency networks.



Research Goal

Compare performance across multiple algorithms:

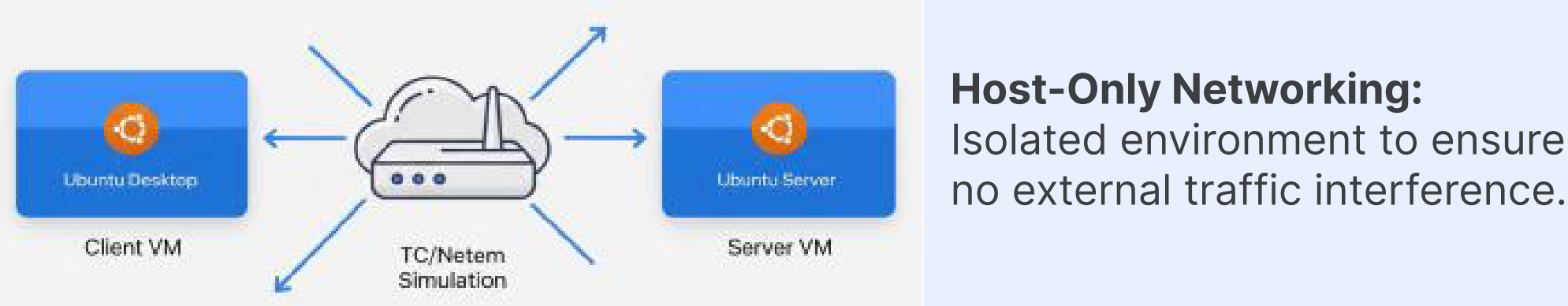
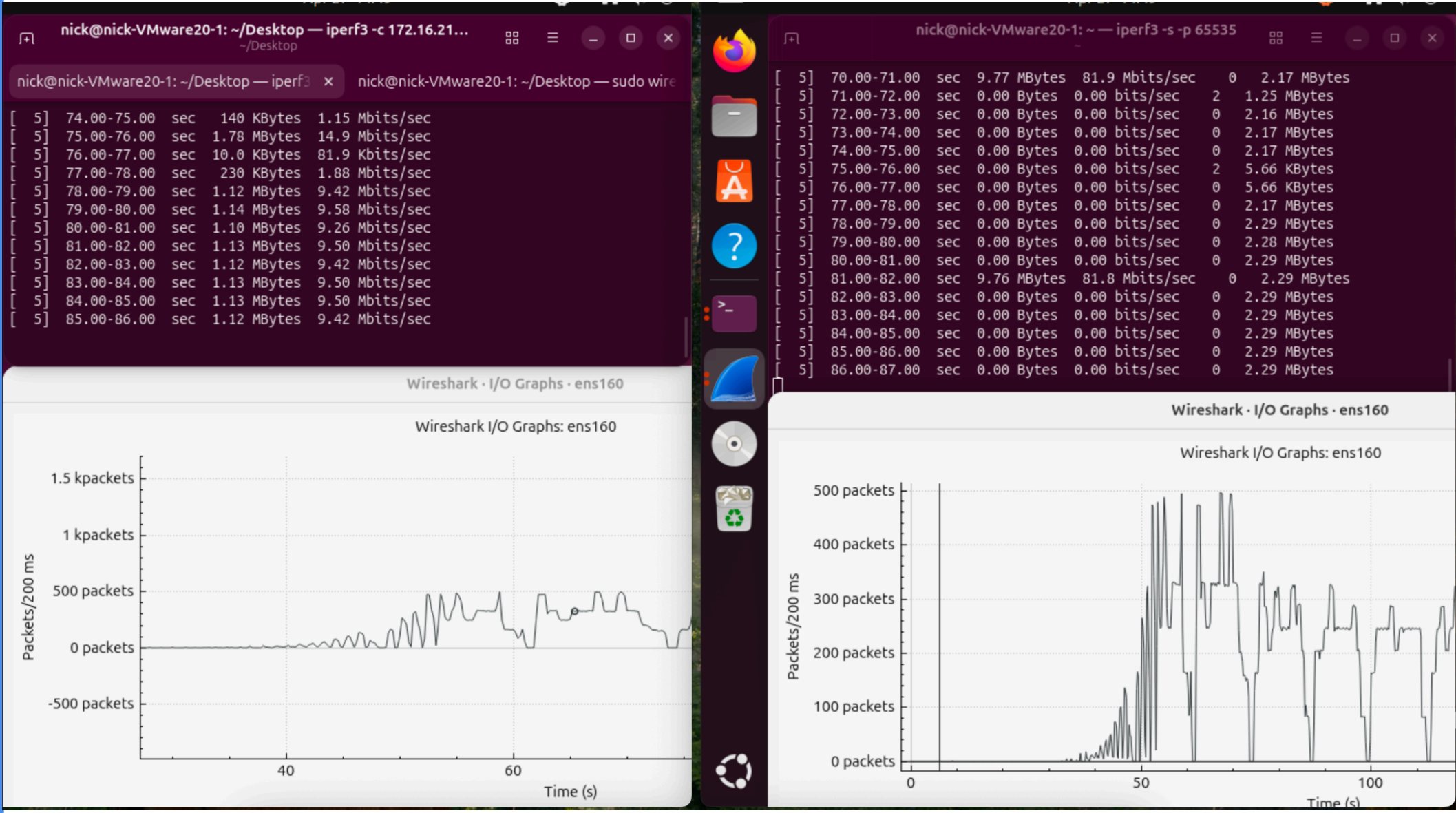
- **BBR3 + CUBIC**

Environment: Ubuntu VM and iperf3

Methodology

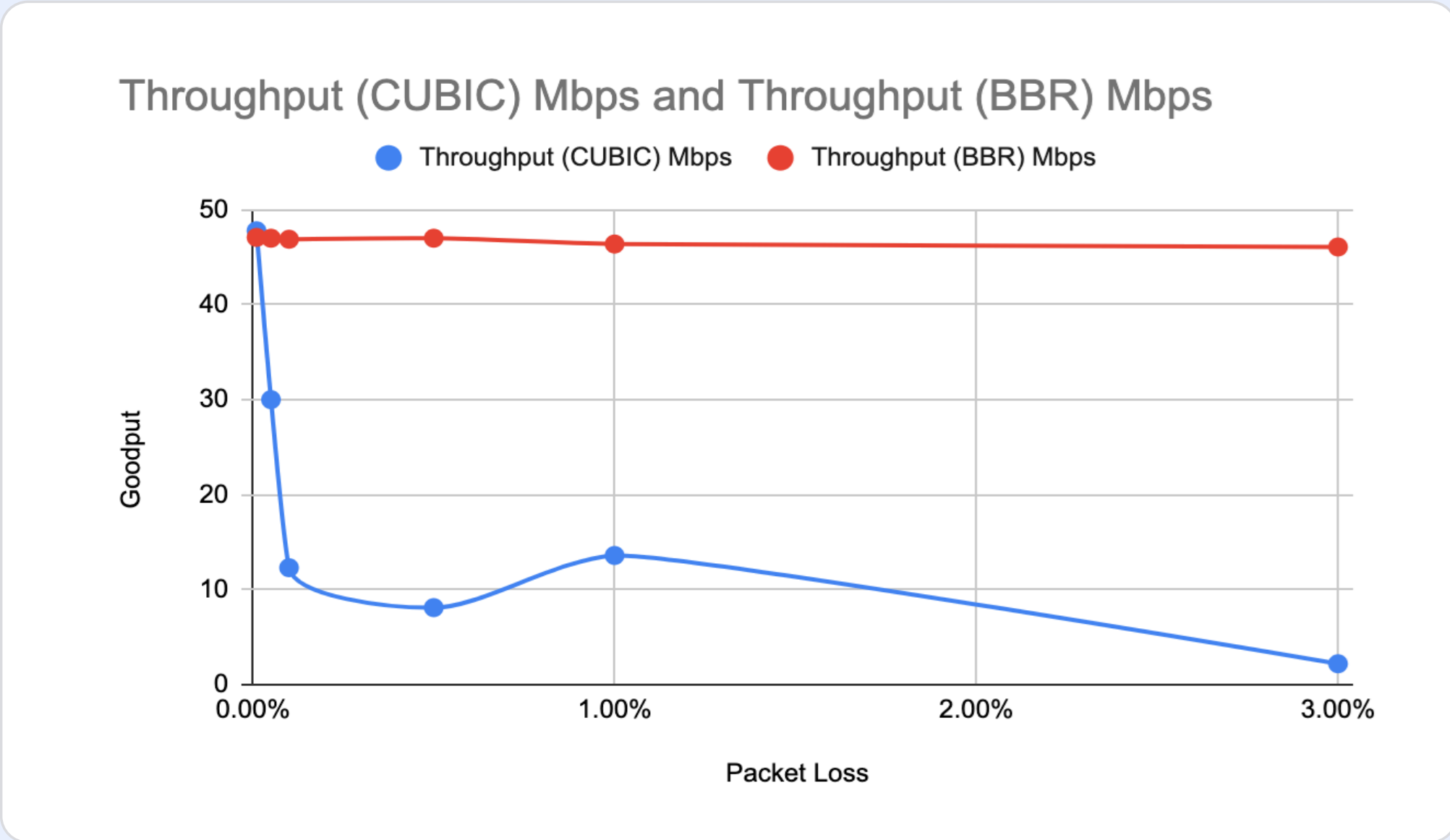
- Setup & Cloning:** Installed Ubuntu 25.10 on VMWare Fusion; created a Linked Clone for client/server pair.
- Link Simulation:** Utilized `tc` and `netem` to tweak latency, jitter, and loss variables.
- Traffic Generation:** Executed `iperf3` tests between VMs to generate traffic using a specific algorithm
- Analysis:** Collected JSON data to interpret trade-offs between throughput and latency control.

Experimental Network Topology



Experimental Results: Packet Loss Analysis

Comparative resilience performance of BBR3 and CUBIC performance under network congestion



Key Observations

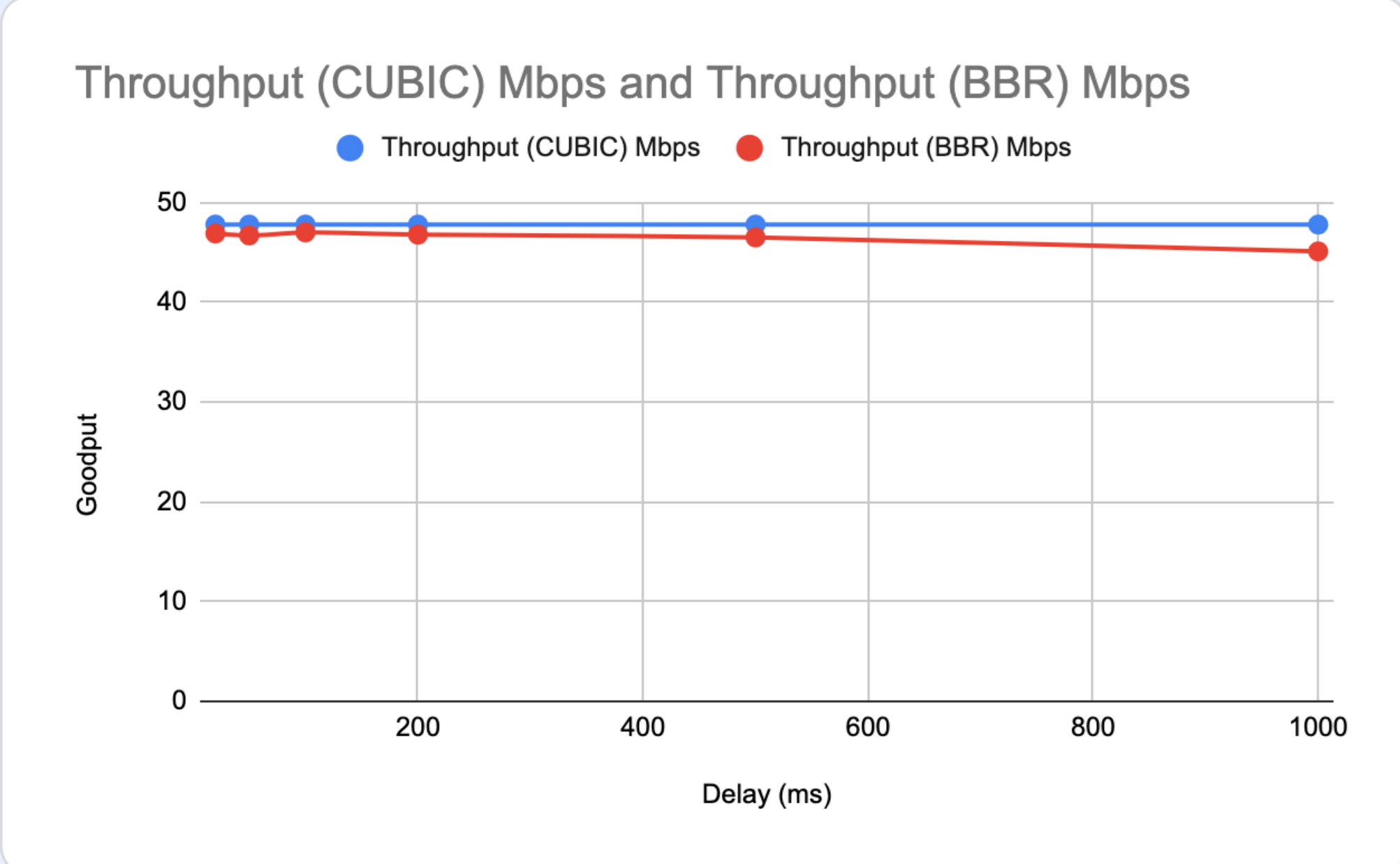
- **BBRv3** maintains a high, stable throughput of ~45 Mbps regardless of packet loss.
- **CUBIC** throughput drops significantly as packet loss increases.

Experiment Conditions:

- **Delay:** 50ms
- **Jitter:** 0ms
- **Loss:** Varied
- **Rate:** 50 Mbps
- **Queue Depth:** 2 × BDP

Experimental Results: Delay Analysis

How CUBIC and BBR respectively respond to greater network delay



Key Observations

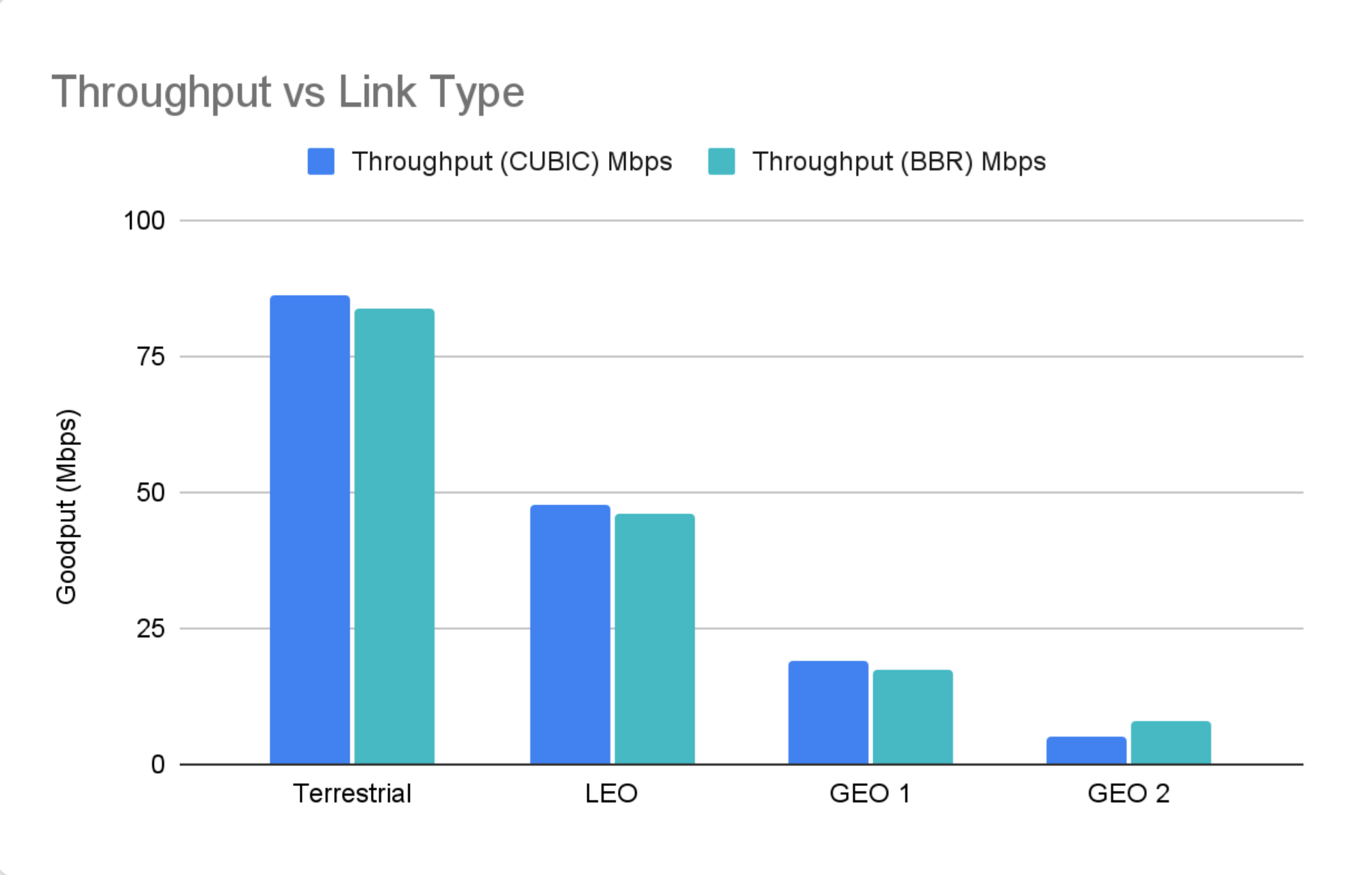
- BBR slightly outperforms CUBIC
- We hypothesize that BBR would outperform CUBIC in shallow buffer conditions

Experiment Conditions:

- **Delay:** Varied
- **Jitter:** 0ms
- **Rate:** 50 Mbps
- **Loss:** 0%
- **Queue Depth:** 2 × BDP

Experimental Results: Common Links

How CUBIC and BBR behave in various representative link conditions



Motivation

Naval vessels rely on LEO/GEO SATCOM with high delay/loss

Experiment Conditions:

- **Loss Correlation:** 25%
- **Queue Depth:** 2 × BDP

Category	Delay (ms)	Jitter (ms)	Rate (Mbps)	Loss
Terrestrial	20	2	90	0.01%
LEO	80	8	50	0.30%
GEO 1	500	50	20	1%
GEO 2	1000	100	10	3%

Future Research:

Test additional TCP algorithms:

- Reno
- Vegas



Additional Test Cases

- Deep vs. Shallow Buffer Conditions
- Additional Congestion Control Algorithms
- Short duration flows
- Multiple competing flows



Investigating Anomalies

- Investigate source of burstiness in iperf
- Investigate periodic stalls in high latency BBRv3 flows